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SEMICONDUCTOR STRUCTURE WITH METAL ION CAPTURE LAYER

Abstract

The present disclosure describes a semiconductor structure with a metal ion capture layer and a method for forming the structure. The method includes forming a first fin structure and a second fin structure on a substrate and forming a first gate structure over the first fin structure and a second gate structure over the second fin structure, where the first gate structure adjoins the second gate structure. The method further includes forming a dielectric layer on the first and second gate structures, removing a portion of the dielectric layer above an adjoining portion of the first and second gate structures to form an opening, and forming a metal ion capture layer in the opening.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a divisional of U.S. Non-Provisional patent application Ser. No. 17/720,481, filed on Apr. 14, 2022, titled “Semiconductor Structure with Metal Ion Capture Layer,” the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] With advances in semiconductor technology, there has been increasing demand for higher storage capacity, faster processing systems, higher performance, and lower costs. To meet this demand, the semiconductor industry continues to scale down the dimensions of semiconductor devices and increase the density of transistors. Metal gates with different work function metals (WFMs) can adjoin each other in the scaled-down semiconductor devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures.

[0004] FIG. 1A illustrates an isometric view of a semiconductor structure with a metal ion capture layer, in accordance with some embodiments.

[0005] FIG. 1B illustrates a cross-sectional view of the semiconductor structure illustrated in FIG. 1A along line A-A, in accordance with some embodiments.

[0006] FIG. 1C illustrates concentration profiles of different ions with respect to width in a semiconductor structure with a metal ion capture layer, in accordance with some embodiments.

[0007] FIG. 1D illustrates an isometric view of another semiconductor structure with a metal ion capture layer, in accordance with some embodiments.

[0008] FIG. 1E illustrates a cross-sectional view of the semiconductor structure illustrated in FIG. 1D along line B-B, in accordance with some embodiments.

[0009] FIG. 1F illustrates concentration profiles of ions formed by different methods with respect to depth in a semiconductor structure with a metal ion capture layer, in accordance with some embodiments.

[0010] FIGS. 1G-1I illustrate metal gate layouts in semiconductor devices, in accordance with some embodiments.

[0011] FIG. 2 is a flow diagram of a method for fabricating a semiconductor structure with a metal ion capture layer, in accordance with some embodiments.

[0012] FIGS. 3-10 illustrate cross-sectional views of a semiconductor structure with a metal ion capture layer at various stages of its fabrication process, in accordance with some embodiments.

[0013] FIG. 11 is a flow diagram of another method for fabricating a semiconductor structure with a metal ion capture layer, in accordance with some embodiments.

[0014] FIGS. 12-18 illustrate cross-sectional views of a semiconductor structure with a metal ion capture layer at various stages of its fabrication process, in accordance with some embodiments.

DETAILED DESCRIPTION

[0015] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the process for forming a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. As used herein, the formation of a first feature on a second feature means the first feature is formed in direct contact with the second feature. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition does not in itself dictate a relationship between the embodiments and/or configurations discussed herein.

[0016] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0017] It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “exemplary,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

[0018] It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.

[0019] In some embodiments, the terms “about” and “substantially” can indicate a value of a given quantity that varies within 5% of the value (e.g., $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$ of the value). These values are merely examples and are not intended to be limiting. The terms “about” and “substantially” can refer to a percentage of the values as interpreted by those skilled in relevant art(s) in light of the teachings herein.

[0020] Fins associated with fin field effect transistors (finFETs) and gate-all-around (GAA) FETs can be patterned by any suitable method. For example, the fins can be patterned using one or more photolithography processes, including double-patterning and multi-patterning processes. Double-patterning and multi-patterning processes can combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, a sacrificial layer can be formed over a substrate and patterned using a photolithography process. Spacers can be formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer can be removed, and the remaining spacers can be used to pattern the fins.

[0021] The discussion of elements in FIGS. 1A, 1B, 1D, 1E, 1G-1I, 3-10, and 12-18 with the same annotations applies to each other, unless mentioned otherwise.

[0022] As the semiconductor industry continues to scale down the dimensions of semiconductor devices, metal gates with different work function metals (WFMs) can be designed to adjoin each other to decrease device footprint. Different WFMs can have different concentrations of the same metal. For example, some WFMs can have a higher concentration of aluminum (Al) or are

aluminum-rich. A WFM alloy can be considered aluminum-rich if the WFM alloy contains more than about 10% of Al in atomic percentage (at %). Some WFMs can have a lower concentration of Al or are Al-free. When WFMs with different concentrations of the same metal adjoin, the metal can diffuse from the higher-concentration side to the lower-concentration side. The metal drifting and/or migration can cause leakage current in the metal gates and reduce device performance.

[0023] The present disclosure provides example FET structures (e.g., GAA FETs, finFETs, or planar FETs) with metal ion capture layers in a semiconductor device and/or in an integrated circuit (IC) and example methods for fabricating the same. In some embodiments, a metal ion capture layer can be formed in an interlayer dielectric (ILD) layer above adjoining metal gates with different WFMs. An opening can be formed in the ILD layer by a dry etch process or a wet etch process. The metal ion capture layer can be formed by depositing a fluorine-rich oxide layer in the opening with a physical vapor deposition (PVD) process or a chemical vapor deposition (CVD) process. An oxide layer can be considered fluorine-rich if the oxide layer contains more than about 10% of fluorine (F) in at %. The F in the fluorine-rich oxide layer can react with metal ions diffused from the higher-concentration side of the WFMs to the lower-concentration side, therefore reducing the metal drifting and/or migration. For example, F and Al can react to form aluminum fluoride (AlF₃). The reaction can take place while the fluorine-rich oxide layer is being deposited. In some embodiments, after the reaction, the fluorine-rich oxide layer can remain in the ILD layer. In some embodiments, the fluorine-rich oxide layer can be removed after the reaction.

[0024] In some embodiments, a metal ion capture layer can be formed in a metal fill layer of adjoining metal gates with different WFMs. In some embodiments, the metal ion capture layer can be formed by diffusing carbon (C), nitrogen (N), phosphorous (P), arsenic (As), antimony (Sb), F, and combinations thereof in the metal fill layer by a thermal diffusion process, an atomic layer deposition (ALD) diffusion process, or a wet diffusion process. In some embodiments, the metal ion capture layer can be formed by implanting C, N, P, As, Sb, F, and combinations thereof in the metal fill layer by an ion implantation process. In some embodiments, the ion implantation process can be plasma enhanced. The area to be diffused or implanted can be defined by a mask layer, such as a photoresist layer. The C, N, P, As, Sb, F, and combinations thereof in the metal ion capture layer can react with metal ions diffused from the higher-concentration side of the WFMs to the lower-concentration side, therefore reducing the metal drifting and/or migration. For example, C, N, P, As, Sb, F, and Al can react to form aluminum compounds. The reaction can take place while the metal ion capture layer is being formed. In some embodiments, after the reaction, the metal ion capture layer can remain in the metal fill layer. In some embodiments, the metal ion capture layer can be removed after the reaction. The metal ion capture layer can reduce metal drifting and/or migration, decrease leakage current in the adjoining metal gates with different WFMs and improve device performance.

[0025] According to some embodiments, FIG. 1A illustrates an isometric view of a FET **100A** and FIG. 1D illustrates an isometric view of a FET **100D**. In some embodiments, FETs **100A** and **100D** can represent a finFET or a GAA FET. In some embodiments, FETs **100A** and **100D** can include n-type FET (NFET) adjoining p-type FET (PFET) and the discussion of FETs **100A** and **100D** applies to both NFET and PFET, unless mentioned otherwise. FIG. 1B illustrates a cross-sectional view of FET **100A** along line A-A showing relative positions between metal ion capture layer **120** and ILD layer **118**. FIG. 1E illustrates a cross-sectional view of FET **100D** along line B-B showing relative positions between metal ion capture layer **121** and metal fill layer **130**. FIGS. 1G-1I illustrate metal gate layouts in semiconductor devices. The discussion of elements in FIGS. 1A, 1B, 1D, 1E, and 1G-1I with the same annotations applies to each other, unless mentioned otherwise.

[0026] Referring to FIGS. 1A and 1D, FETs **100A** and **100D** can be formed on a substrate **104**. There can be other FETs and/or structures (e.g., isolation structures) formed on substrate **104**. Substrate **104** can be a semiconductor material, such as silicon (Si), germanium (Ge), silicon germanium (SiGe), a silicon-on-insulator (SOI) structure, and combinations thereof. Further,

substrate **104** can be doped with p-type dopants, such as boron (B), indium (In), aluminum (Al), and gallium (Ga), or n-type dopants, such as P and As.

[0027] Referring to FIGS. **1A**, **1D**, and **1G-1I**, FETs **100A** and **100D** can include n-type fin structures **106A** and p-type fin structures **106B** disposed on substrate **104**. N-type fin structures **106A** can be doped with n-type dopants, such as P and As. P-type fin structures **106B** can be doped with p-type dopants, such as B, In, Al, and Ga. In some embodiments, n-type fin structures **106A** and p-type fin structures **106B** can include a material similar to substrate **104** and extend along the X-axis. For GAA FETs, n-type fin structures **106A** and p-type fin structures **106B** can include first-type and second-type nanostructures (not shown in FIGS. **1A**, **1D**, and **1G-1I**). The first-type nanostructures can include a semiconductor material similar to substrate **104**. The second-type nanostructures can include a semiconductor material similar to or different from substrate **104**. In some embodiments, the second-type nanostructures can include Si, silicon arsenide (SiAs), silicon phosphide (SiP), silicon carbide (SiC), silicon carbon phosphide (SiCP), SiGe, silicon germanium boron (SiGeB), germanium boron (GeB), silicon germanium stannum boron (SiGeSnB), a III-V semiconductor compound, or other suitable semiconductor materials. The second-type nanostructures can be nanosheets or nanowires. The second-type nanostructures can have cross-sections of other geometric shapes, such as circular, elliptical, triangular, and polygonal shapes.

[0028] Referring to FIGS. **1A** and **1D**, FETs **100A** and **100D** can include p-type source/drain (S/D) regions **110A** disposed on portions of n-type fin structures **106A** and n-type source/drain (S/D) regions **110B** disposed on portions of p-type fin structures **106B**. For PFETs, each of S/D regions **110A** can include an epitaxially-grown semiconductor material, such as Si and SiGe, and p-type dopants, such as B and other suitable p-type dopants. For NFETs, each of S/D regions **110B** can include an epitaxially-grown semiconductor material, such as Si and SiGe, and n-type dopants, such as P and other suitable n-type dopants. A dopant concentration can be in a range from about 1×10^{20} atoms/cm³ to about 3×10^{22} atoms/cm³ in each of S/D regions **110A** and **110B**. In some embodiments, S/D regions **110A** and **110B** can have a depth of about 50 nm to about 70 nm.

[0029] Referring to FIGS. **1A**, **1D**, and **1G-1I**, FETs **100A** and **100D** can include p-type gate structures **112A** disposed on n-type fin structures **106A** and n-type gate structures **112B** (not visible in FIG. **1A**) disposed on p-type fin structures **106B**. P-type gate structures **112A** can have a p-type WFM. A p-type WFM can have a work function between about 4.5 eV and about 5.5 eV. N-type gate structures **112B** can have an n-type WFM. An n-type WFM can have a work function between about 3.5 eV and about 4.5 eV. In some embodiments, gate structures **112A** and **112B** can be multi-layered structures. The multiple layers of gate structures **112A** and **112B** are not shown in FIGS. **1A** and **1D** for simplicity.

[0030] Referring to FIGS. **1B** and **1E**, each of gate structures **112A** and **112B** can include an interfacial oxide (IO) layer (not shown in FIGS. **1B** and **1E**), a high-k (HK) dielectric layer **122** disposed on the IO layer, and a conductive layer disposed on HK dielectric layer **122**. The IO layer can include SiO_x, SiGeO_x, or germanium oxide (GeO_x). HK dielectric layer **122** can include a HK dielectric material, such as hafnium oxide (HfO₂), titanium oxide (TiO₂), hafnium zirconium oxide (HfZrO), tantalum oxide (Ta₂O₃), hafnium silicate (HfSiO₄), zirconium oxide (ZrO₂), and zirconium silicate (ZrSiO₂). HK dielectric layer **122** can have a thickness H₅ between about 1 nm and about 10 nm. The conductive layer can be a multi-layered structure. The conductive layer can include a WFM layer disposed on HK dielectric layer **122** and a metal fill layer **130** on the WFM layer. In some embodiments, n-type gate structures **112B** can include an n-type WFM layer **126**. N-type WFM layer **126** can include titanium aluminum (TiAl), titanium aluminum carbide (TiAlC), tantalum aluminum (TaAl), tantalum aluminum carbide (TaAlC), Al-doped titanium (Ti), Al-doped titanium nitride (TiN), Al-doped tantalum (Ta), Al-doped tantalum nitride (TaN), other suitable Al-based materials, and combinations thereof. N-type WFM layer **126** can include other metal alloys that have an n-type

work function. In some embodiments, p-type gate structures **112A** can include a p-type WFM layer **124**. P-type WFM layer **124** can include substantially Al-free (e.g., with no Al) Ti-based or Ta-based nitrides or alloys, such as TiN, titanium silicon nitride (TiSiN), titanium gold (Ti—Au) alloy, titanium copper (Ti—Cu) alloy, TaN, tantalum silicon nitride (TaSiN), tantalum gold (Ta—Au) alloy, tantalum copper (Ta—Cu), and combinations thereof. P-type WFM layer **124** can include other metal alloys that have a p-type work function. In some embodiments, the WFM layer of p-type gate structures **112A** can include a stack of the same WFM. For example, the WFM layer of p-type gate structures **112A** can include two or more layers of TiN. In some embodiments, the WFM layer of p-type gate structures **112A** can include a stack of different WFMs. For example, the WFM layer of p-type gate structures **112A** can include p-type WFM layer **124** and n-type WFM layer **126** disposed on p-type WFM layer **124**.

[0031] For example, as shown in FIGS. **1B** and **1E**, p-type gate structure **112A** can include a WFM stack including p-type WFM layer **124**, such as TiN, and n-type WFM layer **126**, such as TiAl. N-type gate structure **112B** can include n-type WFM layer **126**, such as TiAl. P-type WFM layer **124** can have a thickness **H6** between about 5 nm and about 50 nm. N-type WFM layer **126** can have a thickness **H4** between about 2 nm and about 30 nm. In some embodiments, p-type gate structure **112A** and n-type gate structure **112B** can include a glue layer **128**, such as TiN. Glue layer **128** can increase adhesion of metal fill layer **130** on the WFMs, such as n-type WFM layer **126**. Glue layer **128** can have a thickness **H3** between about 1 nm and about 10 nm. Metal fill layer **130** can include a suitable conductive material, such as tungsten (W), low-fluorine tungsten (LFW), Ti, silver (Ag), ruthenium (Ru), molybdenum (Mo), copper (Cu), cobalt (Co), Al, iridium (Ir), nickel (Ni), metal alloys, and combinations thereof. Metal fill layer **130** on the side of p-type gate structures **112A** can have a thickness **H7** between about 2 nm and about 40 nm. Metal fill layer **130** on the side of n-type gate structures **112B** can have a thickness **H2** between about 5 nm and about 60 nm.

[0032] Referring to FIGS. **1A** and **1D**, FETs **100A** and **100D** can include gate spacers **114** and shallow trench isolation (STI) regions **116**. FETs **100A** and **100D** can include etch stop layers (ESLs) **117** and ILD layers **118**. ILD layers **118** can be disposed on ESLs **117**. GAA FETs can further include inner spacers (not shown in FIGS. **1A** and **1D**). In some embodiments, gate spacers **114**, inner spacers, STI regions **116**, ESLs **117**, and ILD layers **118** can include an insulating material, such as silicon oxide (SiO.sub.x), silicon nitride (SiN), silicon carbon nitride (SiCN), silicon oxycarbon nitride (SiOCN), and silicon germanium oxide (SiGeO.sub.x). In some embodiments, gate spacers **114** and inner spacers can have a thickness between about 2 nm and about 9 nm for adequate electrical isolation of gate structures **112A** and **112B** from adjacent structures.

[0033] Referring to FIGS. **1A** and **1B**, FET **100A** can include ILD layer **118** above adjoining gate structures **112A** and **112B** (not visible in FIG. **1A**). FET **100A** can include metal ion capture layer **120** disposed within ILD layer **118** at or near the adjoining region of gate structures **112A** and **112B**. Metal ion capture layer **120** can include a fluorine-rich oxide layer, such as fluorine-doped metal oxide (MO.sub.x) and fluorine-doped SiO.sub.x. An oxide layer can be considered fluorine-rich if the oxide layer contains more than about 10% of F in at %. Metal ion capture layer **120** can have a thickness **H1** between about 10 nm and about 20 nm, between about 5 nm and about 30 nm, and between about 2 nm and about 40 nm. If the thickness **H1** is less than about 2 nm, metal ion capture layer **120** cannot effectively reduce the diffused metal ions. Reduction of the diffused metal ions is not effective if the metal ion concentration at or near the adjoining point of the gate structures with metal ion capture layer **120** is more than 75% of the metal ion concentration at or near the adjoining point of the gate structures without metal ion capture layer **120**. If the thickness **H1** is greater than about 40 nm, the manufacturing cost of forming metal ion capture layer **120** can be too high. Metal ion capture layer **120** can have a width **W1** between about 40 nm and about 100 nm, between about 20 nm and about 150 nm, and between about 10 nm and about 200 nm. If the width **W1** is less than about 10 nm, metal ion capture layer **120** cannot effectively reduce the

diffused metal ions. Reduction of the diffused metal ions is not effective if the metal ion concentration at or near the adjoining point of the gate structures with metal ion capture layer **120** is more than 75% of the metal ion concentration at or near the adjoining point of the gate structures without metal ion capture layer **120**. If the width **W1** is greater than about 200 nm, the manufacturing cost of forming metal ion capture layer **120** can be too high.

[0034] Referring to FIG. **1C**, the concentration of **F** with respect to width can be the highest at (or near) the center of metal ion capture layer **120** or at (or near) the center of adjoining point of gate structures **112A** and **112B**. **F** in metal ion capture layer **120** can diffuse toward two sides of metal ion capture layer **120** along the **Y**-axis. The concentration of **F** can decrease as the distance from the center of metal ion capture layer **120** increases. The concentration of **Al** with respect to width can be the highest in n-type WFM layer **126** because it is aluminum-rich **TiAl** and the lowest in p-type WFM layer **124** because it is **Al**-free **TiN**. A WFM alloy can be considered aluminum-rich if the WFM alloy contains more than about 10% of **Al** in at %. **Al** can diffuse and/or migrate from n-type WFM layer **126** to p-type WFM layer **124**. At or near the adjoining point of gate structures **112A** and **112B**, **Al** can react with **F** from metal ion capture layer **120**. For example, **F** and **Al** can react to form **AlF₃**. The reaction can reduce the concentration of **Al**. Because the concentration of **F** is the highest at (or near) the center of metal ion capture layer **120**, the reaction is the strongest. Therefore, the concentration of **Al** can be the lowest at (or near) the center of metal ion capture layer **120**. At or near the center of metal ion capture layer **120**, the concentration of **Al** can be lower than that of n-type WFM layer **126** similarly because of the reaction. The reaction takes place and the concentration of **Al** decreases as metal ion capture layer **120** is being formed. In some embodiments, after the reaction, **ILD** layer **118** and metal ion capture layer **120** can be removed. In some embodiments, unlike as shown in FIG. **1A**, **ILD** layer **118** and metal ion capture layer **120** do not exist in the final FET structures.

[0035] Referring to FIGS. **1D** and **1E**, FET **100D** can include metal ion capture layer **121** disposed within gate structures **112A** and **112B**. As shown in FIG. **1E**, metal ion capture layer **121** can be disposed within metal fill layer **130** at or near the adjoining region of gate structures **112A** and **112B**. Metal ion capture layer **121** can include dopants such as **C**, **N**, **P**, **As**, **Sb**, **F**, and combinations thereof within metal fill layer **130**. Metal ion capture layer **121** can have a thickness **H8** between about 4 nm and about 10 nm, between about 2 nm and about 15 nm, and between about 1 nm and about 20 nm. A ratio between **H8** and **H7** can be between about 0.2 and about 0.3, between about 0.15 and about 0.4, and between about 0.1 and about 0.5. If the thickness **H8** is less than about 1 nm or if the ratio **H8/H7** is less than about 0.1, metal ion capture layer **121** cannot effectively reduce the diffused metal ions. Reduction of the diffused metal ions is not effective if the metal ion concentration at or near the adjoining point of the gate structures with metal ion capture layer **121** is more than 75% of the metal ion concentration at or near the adjoining point of the gate structures without metal ion capture layer **121**. If the thickness **H8** is greater than about 20 nm or if the ratio **H8/H7** is greater than about 0.5, the manufacturing cost of forming metal ion capture layer **121** can be too high. Metal ion capture layer **121** can have a width **W2** between about 40 nm and about 100 nm, between about 20 nm and about 150 nm, and between about 10 nm and about 200 nm. If the width **W2** is less than about 10 nm, metal ion capture layer **121** cannot effectively reduce the diffused metal ions. Reduction of the diffused metal ions is not effective if the metal ion concentration at or near the adjoining point of the gate structures with metal ion capture layer **121** is more than 75% of the metal ion concentration at or near the adjoining point of the gate structures without metal ion capture layer **121**. If the width **W2** is greater than about 200 nm, the manufacturing cost of forming metal ion capture layer **121** can be too high.

[0036] Referring to FIG. **1F**, the concentration of the dopant atoms diffused and/or implanted in metal ion capture layer **121** can increase with depth first and then decrease. The most concentrated point with respect to depth can be shallower with ion implantation and can be deeper with diffusion. The concentration of the dopant atoms implanted at the most concentrated point can be

greater than that diffused at the most concentrated point. The entire depth of implantation can be less than the entire depth of diffusion. In some embodiments, the concentration of the dopant atoms diffused and/or implanted in metal ion capture layer **121** can decrease with depth from the top surface of metal ion capture layer **121**. In some embodiments, the concentration of the dopant atoms diffused and/or implanted in metal ion capture layer **121** can be substantially uniform. The concentration of the dopant atoms in metal ion capture layer **121** can range from about 1×10^{20} atoms/cm³ to about 3×10^{22} atoms/cm³, from about 0.8×10^{20} atoms/cm³ to about 3.3×10^{22} atoms/cm³, and from about 0.5×10^{20} atoms/cm³ to about 3.5×10^{22} atoms/cm³. If the concentration is less than about 0.5×10^{20} atoms/cm³, metal ion capture layer **121** cannot effectively reduce the diffused metal ions. Reduction of the diffused metal ions is not effective if the metal ion concentration at or near the adjoining point of the gate structures with metal ion capture layer **121** is more than 75% of the metal ion concentration at or near the adjoining point of the gate structures without metal ion capture layer **121**. If the concentration is greater than about 3.5×10^{22} atoms/cm³, the manufacturing cost of forming metal ion capture layer **121** can be too high.

[0037] The concentration of the dopant atoms diffused and/or implanted in metal ion capture layer **121** can change with respect to width in a manner similar to the concentration change of F described with reference to FIG. **1C**. The concentration of Al at or near the adjoining point of gate structures **112A** and **112B** can change with respect to width in a manner similar to the concentration change of Al described with reference to FIG. **1C**, because Al can react with the dopant atoms from metal ion capture layer **121**. For example, the dopant atoms and Al can react to form aluminum compounds. The reaction takes place and the concentration of Al decreases as metal ion capture layer **121** is being formed. In some embodiments, after the reaction, metal ion capture layer **121** can be removed. In some embodiments, unlike as shown in FIG. **1D**, metal ion capture layer **121** does not exist in the final FET structures.

[0038] FIG. **1G** illustrates a metal gate layout design in a semiconductor device. The semiconductor device can include n-type fin structures **106A** and p-type fin structures **106B**. The semiconductor device can include p-type gate structures **112A** disposed on n-type fin structures **106A** and n-type gate structures **112B** disposed on p-type fin structures **106B**. Area A illustrates the adjoining regions of p-type gate structures **112A** and n-type gate structures **112B**. Area A can include metal ion capture layers **120** or **121** (not shown in FIG. **1G**). In some embodiments, the semiconductor device illustrated in FIG. **1G** can be a ring oscillator. The ring oscillator can be used to provide a clock signal and benchmark device speed and power standby.

[0039] FIG. **1H** illustrates a metal gate layout design in a semiconductor device. The semiconductor device can include n-type fin structures **106A** and p-type fin structures **106B**. The semiconductor device can include p-type gate structures **112A** disposed on n-type fin structures **106A** and n-type gate structures **112B** disposed on p-type fin structures **106B**. The semiconductor device can include cut metal gates **132**. Cut metal gates **132** can include isolation structures between adjacent gate structures and therefore do not need metal ion capture layers **120** or **121**. Area B illustrates the adjoining regions of p-type gate structures **112A** and n-type gate structures **112B**. Area B can include metal ion capture layers **120** or **121** (not shown in FIG. **1H**). In some embodiments, the semiconductor device illustrated in FIG. **1H** can be a 6-transistor (6T) static random access memory (SRAM). The 6T SRAM can be used to hold, read, and write data. The 6T SRAM can include pull-up (PU) transistors, pull-down (PD) transistors, and pass-gate (PG) transistors.

[0040] FIG. **1I** illustrates a metal gate layout design in a semiconductor device. The semiconductor device can include n-type fin structures **106A** and p-type fin structures **106B**. The semiconductor device can include p-type gate structures **112A** disposed on n-type fin structures **106A** and n-type gate structures **112B** disposed on p-type fin structures **106B**. The semiconductor device can include cut metal gates **132**. Cut metal gates **132** can include isolation structures between adjacent gate

structures and therefore do not need metal ion capture layers **120** or **121**. Areas C, D, E, and F illustrate the adjoining regions of p-type gate structures **112A** and n-type gate structures **112B**. Areas C, D, E, and F can include metal ion capture layers **120** or **121** (not shown in FIG. **1I**). FIG. **1I** illustrates the versatility of forming metal ion capture layers **120** or **121**. A photolithography process can be used to pinpoint the adjoining regions where metal ion capture layers **120** or **121** are needed. Metal ion capture layers **120** or **121** can be formed at or near the adjoining regions of gate structures with different WFMs. Metal ion capture layers **120** or **121** can reduce leakage current caused by WFM ion drifting and/or migration and improve device performance.

[0041] According to some embodiments, FIG. **2** is a flow diagram describing a method **200** for fabricating FET **100A**, as shown in FIGS. **1A** and **1B**. For illustrative purposes, the operations illustrated in FIG. **2** will be described with reference to the example fabrication process for fabricating FET **100A** as illustrated in FIGS. **3-10**. FIGS. **3-10** are cross-sectional views of FET **100A** along line A-A of FIG. **1A** at various stages of fabrication, according to some embodiments. Additional fabrication operations can be performed between the various operations of method **200** and are omitted for simplicity. These additional fabrication operations are within the spirit and the scope of this disclosure. Moreover, not all operations may be required to perform the disclosure provided herein. Additionally, some of the operations can be performed simultaneously or in a different order than the ones shown in FIG. **2**. Elements in FIGS. **3-10** with the same annotations as elements in FIGS. **1A** and **1B** are described above.

[0042] It should be noted that method **200** may not produce a complete FET **100A**. Accordingly, it is understood that additional processes can be provided before, during, and after method **200**, and that some other processes may only be briefly described herein. For example, in some embodiments, fin structures can be patterned by a photolithography process on a semiconductor substrate. Polysilicon structures can be deposited by a CVD process over the fin structures and patterned by a photolithography process. Spacers can be deposited by a CVD process on the polysilicon structures. Portions of the fin structures can be removed by a dry etch process to form S/D openings, and S/D regions can be epitaxially grown in the S/D openings. In GAA FETs, inner spacers can be deposited by a CVD process prior to the S/D region formation. The polysilicon structures can then be removed by a dry etch process to form gate openings and metal gate structures can be formed in the gate openings.

[0043] Referring to FIG. **2**, in operation **202**, a HK dielectric layer is deposited in a first gate opening and a second gate opening. For example, as shown in FIG. **3**, HK dielectric layer **122** can be formed in first gate opening **302** and second gate opening **304**. First and second gate openings **302** and **304** are not shown with sidewalls in FIG. **3**. First and second gate openings **302** and **304** adjoin each other. In some embodiments, first gate opening **302** can be used to form a p-type metal gate with a p-type WFM and second gate opening **304** can be used to form an n-type metal gate with an n-type WFM. HK dielectric layer **122** can be deposited by a CVD process or a PVD process. In some embodiments, an IO layer (not shown in FIG. **3**) can be formed by a CVD process or a PVD process in first and second gate openings **302** and **304**.

[0044] Referring to FIG. **2**, in operation **204**, a first-type WFM layer is formed on the HK dielectric layer in the first gate opening. For example, as shown in FIG. **4**, p-type WFM layer **124** can be formed on HK dielectric layer **122** in first gate opening **302**. P-type WFM layer **124** can be selectively formed in first gate opening **302** and not in second gate opening **304** by a photolithography patterning process. P-type WFM layer **124** can be formed by a CVD process, a PVD process, or a metal-organic chemical vapor deposition (MOCVD) process.

[0045] Referring to FIG. **2**, in operation **206**, a second-type WFM layer is formed on the HK dielectric layer in the second gate opening. In some embodiments, the second-type WFM layer is formed on the HK dielectric layer both in the first and second gate openings. For example, as shown in FIG. **5**, n-type WFM layer **126** can be formed on p-type WFM layer **124** in first gate opening **302** and on HK dielectric layer **122** in second gate opening **304**. N-type WFM layer **126**

can be formed by a CVD process, a PVD process, or a MOCVD process.

[0046] Referring to FIG. 2, in operation **208**, a glue layer is formed on the first-type WFM layer and the second-type WFM layer. In some embodiments, the glue layer is formed on the second-type WFM layer where the second-type WFM layer covers the first-type WFM layer. For example, as shown in FIG. 6, glue layer **128** can be formed on n-type WFM layer **126** in first and second gate openings **302** and **304**. Glue layer **128** can be formed by a CVD process, a PVD process, a MOCVD process, or an ALD process.

[0047] Referring to FIG. 2, in operation **210**, a metal fill layer is formed on the glue layer to form a first gate structure and a second gate structure. For example, as shown in FIG. 7, metal fill layer **130** can be formed on glue layer **128** in first and second gate openings **302** and **304** to form first gate structure **112A** and second gate structures **112B**. First and second gate structures **112A** and **112B** adjoin each other. In some embodiments, first gate structure **112A** can be a p-type metal gate with p-type WFM layer **124** and second gate structure **112B** can be an n-type metal gate with n-type WFM layer **126**. Metal fill layer **130** can be formed by a CVD process, a PVD process, or a MOCVD process.

[0048] Referring to FIG. 2, in operation **212**, a dielectric layer is formed on the first and second gate structures. For example, as shown in FIG. 8, ILD layer **118** can be formed on metal fill layer **130** on first and second gate structures **112A** and **112B**. ILD layer **118** can be formed by a CVD process or a PVD process.

[0049] Referring to FIG. 2, in operation **214**, a portion of the dielectric layer above an adjoining portion of the first and second gate structures is removed to form an opening. For example, as shown in FIG. 9, opening **902** can be formed in ILD layer **118** to expose an adjoining region of first and second gate structures **112A** and **112B**. In some embodiments, opening **902** can be formed by a dry etch process (e.g., reactive ion etch process) using fluorocarbon (C.sub.xF.sub.y) gas having a flow rate ranging from about 100 sccm to about 400 sccm. The etch process can be carried out for a period of time ranging from about 10 seconds to about 90 seconds at a temperature ranging from about 10° C. to about 90° C., under a pressure ranging from about 15 mTorr to about 100 mTorr. The etch conditions can depend on the size of opening **902**. The etch process can have a higher etch rate in the vertical direction (e.g., Z-direction) than in the horizontal direction (e.g., Y-direction). The etch process can have a higher etch rate near the top of opening **902** than near the bottom of opening **902**. In some embodiments, opening **902** can be formed by a wet etch process using hydrogen peroxide at a temperature ranging from about 30° C. to about 100° C. In some embodiments, the wet etch process can include a diluted solution of hydrogen fluoride (HF) with a buffer, such as ammonium fluoride (NH.sub.4F), diluted HF (HF/H.sub.2O), phosphoric acid (H.sub.3PO.sub.4), sulfuric acid with deionized water (H.sub.2SO.sub.4/H.sub.2O), and combinations thereof. The etch process can be a timed etch.

[0050] Referring to FIG. 2, in operation **216**, a metal ion capture layer is formed in the opening. For example, as shown in FIG. 10, metal ion capture layer **120** can be formed in opening **902** above the adjoining region between first and second gate structures **112A** and **112B**. Metal ion capture layer **120** can be formed by a CVD process. In some embodiments, the CVD process can use a precursor mixture of a metal-organic compound, oxygen, fluorine or a fluorine-containing compound, and an inert carrier gas. In some embodiments, the CVD process can use a precursor mixture of hexachlorodisilane (HCDS), oxygen, fluorine or a fluorine-containing compound, and an inert carrier gas. In some embodiments, the CVD process can be carried out for a period of time ranging from about 10 seconds to about 150 seconds at a temperature ranging from about 20° C. to about 200° C., under a pressure ranging from about 30 Torr to about 150 Torr. In some embodiments, the CVD process can be carried out in atmospheric pressure. If the deposition time is less than about 10 seconds, or the temperature is less than about 20° C., or the pressure is less than about 30 Torr, metal ion capture layer **120** deposited can be too thin and cannot effectively reduce the diffused metal ions. Reduction of the diffused metal ions is not effective if the metal ion

concentration at or near the adjoining point of the gate structures with metal ion capture layer **120** is more than 75% of the metal ion concentration at or near the adjoining point of the gate structures without metal ion capture layer **120**. If the deposition time is greater than about 150 seconds, or the temperature is greater than about 200° C., or the pressure is greater than about 150 Torr, the manufacturing cost of forming metal ion capture layer **120** can be too high. Metal ion capture layers **120** can reduce leakage current caused by WFM ion drifting and/or migration and improve device performance.

[0051] According to some embodiments, FIG. **11** is a flow diagram describing a method **1100** for fabricating FET **100D**, as shown in FIGS. **1D** and **1E**. For illustrative purposes, the operations illustrated in FIG. **11** will be described with reference to the example fabrication process for fabricating FET **100D** as illustrated in FIGS. **12-18**. FIGS. **12-18** are cross-sectional views of FET **100D** along line B-B of FIG. **1D** at various stages of fabrication, according to some embodiments. Additional fabrication operations can be performed between the various operations of method **1100** and are omitted for simplicity. These additional fabrication operations are within the spirit and the scope of this disclosure. Moreover, not all operations may be required to perform the disclosure provided herein. Additionally, some of the operations can be performed simultaneously or in a different order than the ones shown in FIG. **11**. Elements in FIGS. **12-18** with the same annotations as elements in FIGS. **1D** and **1E** are described above. It should be noted that method **1100** may not produce a complete FET **100D**. Accordingly, it is understood that additional processes can be provided before, during, and after method **1100**, and that some other processes may only be briefly described herein.

[0052] Referring to FIG. **11**, in operation **1102**, a HK dielectric layer is deposited in a first gate opening and a second gate opening. For example, as shown in FIG. **12**, HK dielectric layer **122** can be formed in first gate opening **302** and second gate opening **304**. First and second gate openings **302** and **304** are not shown with sidewalls in FIG. **12**. HK dielectric layer **122** can be formed in a manner similar to that described with reference to FIG. **3** and operation **202** of FIG. **2**.

[0053] Referring to FIG. **11**, in operation **1104**, a first-type WFM layer is formed on the HK dielectric layer in the first gate opening. For example, as shown in FIG. **13**, p-type WFM layer **124** can be formed on HK dielectric layer **122** in first gate opening **302**. P-type WFM layer **124** can be formed in a manner similar to that described with reference to FIG. **4** and operation **204** of FIG. **2**.

[0054] Referring to FIG. **11**, in operation **1106**, a second-type WFM layer is formed on the HK dielectric layer in the second gate opening. In some embodiments, the second-type WFM layer is formed on the HK dielectric layer both in the first and second gate openings. For example, as shown in FIG. **14**, n-type WFM layer **126** can be formed on p-type WFM layer **124** in first gate opening **302** and on HK dielectric layer **122** in second gate opening **304**. N-type WFM layer **126** can be formed in a manner similar to that described with reference to FIG. **5** and operation **206** of FIG. **2**.

[0055] Referring to FIG. **11**, in operation **1108**, a glue layer is formed on the first-type WFM layer and the second-type WFM layer. In some embodiments, the glue layer is formed on the second-type WFM layer where the second-type WFM layer covers the first-type WFM layer. For example, as shown in FIG. **15**, glue layer **128** can be formed on n-type WFM layer **126** in first and second gate openings **302** and **304**. Glue layer **128** can be formed in a manner similar to that described with reference to FIG. **6** and operation **208** of FIG. **2**.

[0056] Referring to FIG. **11**, in operation **1110**, a metal fill layer is formed on the glue layer to form a first gate structure and a second gate structure. For example, as shown in FIG. **16**, metal fill layer **130** can be formed on glue layer **128** in first and second gate openings **302** and **304** to form first gate structure **112A** and second gate structure **112B**. First and second gate structures **112A** and **112B** adjoin each other. In some embodiments, first gate structure **112A** can be a p-type metal gate with p-type WFM layer **124** and second gate structure **112B** can be an n-type metal gate with n-type WFM layer **126**. Metal fill layer **130** can be formed in a manner similar to that described with

reference to FIG. 7 and operation **210** of FIG. 2.

[0057] Referring to FIG. **11**, in operation **1112**, a metal ion capture layer is formed in a top portion of the metal fill layer above an adjoining portion of the first and second gate structures. For example, as shown in FIG. **17**, a photoresist layer **1702** can be spin coated on metal fill layer **130**. An opening **1704** above an adjoining region of first and second gate structures **112A** and **112B** can be formed by transferring a pattern of a photomask, such as a reticle, to photoresist layer **1702** by an exposure process and a develop process. Opening **1704** can be exposed to a thermal diffusion process, an ALD diffusion process, a wet diffusion process, or an ion implantation process. In some embodiments, the ion implantation process can be plasma enhanced. As shown in FIG. **18**, after the diffusion process or the ion implantation process, metal ion capture layer **121** can be formed in a top portion of metal fill layer **130** above the adjoining region of first and second gate structures **112A** and **112B**.

[0058] In some embodiments, metal ion capture layer **121** can be formed by a thermal diffusion process. In some embodiments, the thermal diffusion process can use a dopant gas containing C, N, P, As, Sb, F, and combinations thereof. For example, the dopant gas can include arsine (AsH.sub.3), phosphine (PH.sub.3), sulfur hexafluoride (SF.sub.6), bromotrifluoromethane (CBrF.sub.3), and combinations thereof. In some embodiments, the thermal diffusion process can be carried out at a temperature ranging from about 50° C. to about 300° C., from about 30° C. to about 400° C., and from about 20° C. to about 500° C. If the temperature is less than about 20° C., metal ion capture layer **121** diffused into metal fill layer **130** can be too thin and cannot effectively reduce the diffused metal ions. Reduction of the diffused metal ions is not effective if the metal ion concentration at or near the adjoining point of the gate structures with metal ion capture layer **121** is more than 75% of the metal ion concentration at or near the adjoining point of the gate structures without metal ion capture layer **121**. If the temperature is greater than about 500° C., the manufacturing cost of forming metal ion capture layer **121** can be too high. In some embodiments, metal ion capture layer **121** can be formed by an ALD diffusion process or a wet diffusion process.

[0059] In some embodiments, metal ion capture layer **121** can be formed by an ion implantation process. In some embodiments, the ion implantation process can include dopant species containing C, N, P, As, Sb, F, and combinations thereof. In some embodiments, the ion implantation process can use an incident angle of about 0°. The ion beam energy can be between about 1 keV and about 10 keV, between about 0.8 keV and about 12 keV, and between about 0.5 keV and about 15 keV. The dose of the dopant species can be between about 1×10^{12} ions/cm² and about 1×10^{15} ions/cm², between about 0.8×10^{12} ions/cm² and about 1.2×10^{15} ions/cm², and between about 0.5×10^{12} ions/cm² and about 1.5×10^{15} ions/cm². If the ion beam energy is less than about 0.5 keV or the dose of the dopant species is less than about 0.5×10^{12} ions/cm², metal ion capture layer **121** implanted in metal fill layer **130** can be too thin and cannot effectively reduce the diffused metal ions. Reduction of the diffused metal ions is not effective if the metal ion concentration at or near the adjoining point of the gate structures with metal ion capture layer **121** is more than 75% of the metal ion concentration at or near the adjoining point of the gate structures without metal ion capture layer **121**. If the ion beam energy is greater than about 15 keV or the dose of the dopant species is greater than about 1.5×10^{15} ions/cm², the manufacturing cost of forming metal ion capture layer **121** can be too high.

[0060] In some embodiments, the ion implantation process can be plasma enhanced. The enhanced ion implantation process can be carried out with a power ranging from about 500 V to about 1000 V, under a pressure ranging from about 1 mTorr to about 100 mTorr. If the power is less than about 500 V or the pressure is less than about 1 mTorr, metal ion capture layer **121** implanted in metal fill layer **130** can be too thin and cannot effectively reduce the diffused metal ions. Reduction of the diffused metal ions is not effective if the metal ion concentration at or near the adjoining point of the gate structures with metal ion capture layer **121** is more than 75% of the metal ion

concentration at or near the adjoining point of the gate structures without metal ion capture layer **121**. If the power is greater than about 1000 V or the pressure is greater than about 100 mTorr, the manufacturing cost of forming metal ion capture layer **121** can be too high. Metal ion capture layers **121** can reduce leakage current caused by WFM ion drifting and/or migration and improve device performance.

[0061] The present disclosure provides example FET structures (e.g., FETs **100A** and **100D**, GAA FETs, finFETs, or planar FETs) with metal ion capture layers (e.g., metal ion capture layers **120** and **121**) in a semiconductor device and/or in an IC and example methods (e.g., methods **200** and **1100**) for fabricating the same. In some embodiments, a metal ion capture layer (e.g., metal ion capture layer **120**) can be formed in an interlayer dielectric (ILD) layer (e.g., ILD layer **118**) above adjoining metal gates with different WFMs. An opening (e.g., opening **902**) can be formed in the ILD layer by a dry etch process or a wet etch process. The metal ion capture layer can be formed by depositing a fluorine-rich oxide layer in the opening with a physical vapor deposition (PVD) process or a chemical vapor deposition (CVD) process. An oxide layer can be considered fluorine-rich if the oxide layer contains more than about 10% of F in at %. The fluorine in the fluorine-rich oxide layer can react with metal ions diffused from the higher-concentration side of the WFMs to the lower-concentration side, therefore reducing the metal drifting and/or migration. For example, fluorine (F) and aluminum (Al) can react to form aluminum fluoride (AlF_3). The reaction can take place while the fluorine-rich oxide layer is being deposited. In some embodiments, after the reaction, the fluorine-rich oxide layer can remain in the ILD layer. In some embodiments, the fluorine-rich oxide layer can be removed after the reaction.

[0062] In some embodiments, a metal ion capture layer (e.g., metal ion capture layer **121**) can be formed in a metal fill layer (e.g., metal fill layer **130**) of adjoining metal gates with different WFMs. In some embodiments, the metal ion capture layer can be formed by diffusing C, N, P, As, Sb, F, and combinations thereof in the metal fill layer by a thermal diffusion process, an ALD diffusion process, or a wet diffusion process. In some embodiments, the metal ion capture layer can be formed by implanting C, N, P, As, Sb, F, and combinations thereof in the metal fill layer by an ion implantation process. In some embodiments, the ion implantation process can be plasma enhanced. The area to be diffused or implanted can be defined by a photoresist layer (e.g., photoresist layer **1702**). The C, N, P, As, Sb, F, and combinations thereof in the metal ion capture layer can react with metal ions diffused from the higher-concentration side of the WFMs to the lower-concentration side, therefore reducing the metal drifting and/or migration. For example, C, N, P, As, Sb, F, and Al can react to form aluminum compounds. The reaction can take place while the metal ion capture layer is being formed. In some embodiments, after the reaction, the metal ion capture layer can remain in the metal fill layer. In some embodiments, the metal ion capture layer can be removed after the reaction. The metal ion capture layer can reduce metal drifting and/or migration, decrease leakage current in the adjoining metal gates with different WFMs, and improve device performance.

[0063] In some embodiments, a method includes forming a first fin structure and a second fin structure on a substrate and forming a first gate structure over the first fin structure and a second gate structure over the second fin structure, where the first gate structure adjoins the second gate structure. The method further includes forming a dielectric layer on the first and second gate structures, removing a portion of the dielectric layer above an adjoining portion of the first and second gate structures to form an opening, and forming a metal ion capture layer in the opening.

[0064] In some embodiments, a method includes forming a first gate structure, where forming the first gate structure includes forming a first-type work function metal (WFM) layer and a metal fill layer. The method further includes forming a second gate structure adjoining the first gate structure, where forming the second gate structure includes forming a second-type WFM layer and the metal fill layer, and where the second-type WFM layer is different from the first-type WFM layer. The method further includes forming a mask layer with an opening above an adjoining portion of the

first and second gate structures, forming a metal ion capture layer in a top portion of the metal fill layer exposed by the opening, and removing the mask layer.

[0065] In some embodiments, a structure includes a first fin structure and a second fin structure on a substrate and a first gate structure over the first fin structure, the first gate structure including a first-type work function metal (WFM) layer and a metal fill layer. The structure further includes a second gate structure over the second fin structure and adjoining the first gate structure, the second gate structure including a second-type WFM layer, where the second-type WFM layer is different from the first-type WFM layer, and the metal fill layer. The structure further includes a metal ion capture layer in a top portion of the metal fill layer above an adjoining portion of the first and second gate structures.

[0066] It is to be appreciated that the Detailed Description section, and not the Abstract of the Disclosure section, is intended to be used to interpret the claims. The Abstract of the Disclosure section may set forth one or more but not all possible embodiments of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the subjoined claims in any way.

[0067] The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art will appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art will also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A structure, comprising: a first fin structure and a second fin structure on a substrate; a first gate structure over the first fin structure, the first gate structure comprising: a first-type work function metal (WFM) layer; and a metal fill layer; a second gate structure over the second fin structure and adjoining the first gate structure, the second gate structure comprising: a second-type WFM layer, wherein the second-type WFM layer is different from the first-type WFM layer; and the metal fill layer; and a metal ion layer in a top portion of the metal fill layer above an adjoining portion of the first and second gate structures.
2. The structure of claim 1, wherein the metal ion layer comprises carbon (C), nitrogen (N), phosphorous (P), arsenic (As), antimony (Sb), fluorine (F), and combinations thereof.
3. The structure of claim 1, wherein the metal ion layer has a width of at least about 50 nm.
4. The structure of claim 1, wherein a first aluminum concentration in the adjoining portion of the first and second gate structures is lower than a second aluminum concentration in the second-type WFM layer of the second gate structure.
5. The structure of claim 1, wherein the first and second gate structures comprise an interfacial oxide (IO) layer and a high-k (HK) dielectric layer on the IO layer.
6. The structure of claim 1, wherein the first and second gate structures comprise a glue layer on the first-type WFM layer and the second-type WFM layer.
7. The structure of claim 6, wherein the glue layer is between the first-type WFM layer and the metal fill layer.
8. The structure of claim 1, wherein the first-type WFM layer comprises an aluminum-free WFM layer, and wherein the second-type WFM layer comprises an aluminum-rich WFM layer.
9. The structure of claim 1, wherein the metal ion layer comprises a fluorine-rich oxide layer with a width of at least about 50 nm.
10. The structure of claim 1, wherein the first-type WFM layer has a width of at least about 30 nm,

and wherein the second-type WFM layer has a width of at least about 50 nm.

11. A structure, comprising: a first gate structure, wherein the first gate structure comprises a first-type work function metal (WFM) layer and a metal fill layer; a second gate structure adjoining the first gate structure, wherein the second gate structure comprises a second-type WFM layer and the metal fill layer, and wherein the second-type WFM layer is different from the first-type WFM layer; and a metal ion layer in a top portion of the metal fill layer.

12. The structure of claim 11, wherein the metal ion layer comprises carbon (C), nitrogen (N), phosphorous (P), arsenic (As), antimony (Sb), fluorine (F), and combinations thereof.

13. The structure of claim 11, wherein the metal ion layer has a width of at least about 50 nm.

14. The structure of claim 11, wherein the first and second gate structures comprise a glue layer on the first-type WFM layer and the second-type WFM layer.

15. The structure of claim 14, wherein the glue layer is between the first-type WFM layer and the metal fill layer.

16. The structure of claim 11, wherein the first and second gate structures comprise an interfacial oxide (IO) layer and a high-k (HK) dielectric layer on the IO layer.

17. The structure of claim 11, wherein the first-type WFM layer has a width of at least about 30 nm, and wherein the second-type WFM layer has a width of at least about 50 nm.

18. A structure, comprising: a first fin structure and a second fin structure on a substrate; a first gate structure over the first fin structure and a second gate structure over the second fin structure; a dielectric layer on the first and second gate structures; and a metal ion layer in the dielectric layer.

19. The structure of claim 18, wherein the first and second gate structures comprise an interfacial oxide (IO) layer and a high-k (HK) dielectric layer on the IO layer.

20. The structure of claim 18, wherein the first gate structure comprises a first-type work function metal (WFM) layer and a metal fill layer, and wherein the second gate structure comprises a second-type WFM layer different from the first-type WFM layer and the metal fill layer.
